BENCHMARK DATASET WHITE PAPER

IO-VNBD Comprehensive Repository, Coordinate Matrix & Metrics Analysis

Inertial and Odometry Vehicle Navigation Benchmark Dataset

Uche Onyekpe · Vasile Palade · Stratis Kanarachos · Alicja Szkolnik
Coventry University, United Kingdom

~100 hrs DRIVING TIME	**~5,700 km** DISTANCE	**5.87M** TELEMETRY ROWS
273 CSV FILES	**~1.11 GB** DATA VOLUME	

§ Executive Summary

The Inertial and Odometry Vehicle Navigation Benchmark Dataset (IO-VNBD) is a pioneering, large-scale, information-rich public benchmark repository created by researchers Uche Onyekpe, Vasile Palade, Stratis Kanarachos, and Alicja Szkolnik at Coventry University, UK.

The repository provides high-resolution ego-motion telemetry designed to advance research in deep learning, inertial odometry, sensor fusion (INS/GPS integration), vehicle dead reckoning, and autonomous vehicle tracking under complex environmental and motion conditions (including GPS-deprived environments).

IO-VNBD DATASET METRICS OVERVIEW

Total Recorded Driving Time	**~100 Hours (~40 hrs Vehicle ECU + ~58 hrs Smartphone)**
Total Driving Distance	**~5,700 km (~1,300 km ECU + ~4,400 km Smartphone)**
Total Telemetry Records	**5,872,306 Rows (5.87M Samples)**
Total Data Files & Volume	**273 CSV Files, ~1.11 GB**
Primary Sampling Frequency	**10 Hz (ECU at 0.10s period; Smartphone IMU at 10 Hz)**
Data Recording Locations	**United Kingdom (England), France, Nigeria**
Participating Drivers	**8 Drivers (A through H; Aggressive & Defensive Styles)**
Test Vehicle Models	**Ford Fiesta Titanium, Volvo XC70, Renault Megane, Toyota Corolla Verso**
Tyre Pressure Scenarios	**5 Standardized Configurations (A, B, C, D, E)**
Driving Scenarios	**32 Distinct Environmental & Dynamic Scenarios**

1. Repository Directory Architecture & Data Breakdown

The repository is structured into two main primary dataset categories: **Synchronised** and **Unsynchronised**, each sub-divided into **Categorised** (grouped by driver and route) and **Uncategorised** datasets.

```
IO-VNBD/
├── README.md                              # Core repository summary documentation
├── README_1.pdf                            # Full peer-reviewed academic publication paper
├── Synchronised V abd S datasets/         # Manually time-aligned vehicle ECU & smartphone pairs
│   ├── Categorised IOVNB Dataset/         # Organized by Driver & Route ID (Drivers A, B, C, D, E)
│   │   ├── M (Driver B)/                  # Driver B: Route M (V-M.csv, S-M.csv)
│   │   ├── S (Driver A)/                  # Driver A: Routes S1, S2, S3a-c, S4
│   │   ├── Vf (Driver E)/                 # Driver E: Route Vfa01-02, Vfb01-02
│   │   ├── Vta (Driver E)/                # Driver E: Route Vta01-Vta30
│   │   ├── Vtb (Driver E)/                # Driver E: Route Vtb01-Vtb13
│   │   ├── Vw (Driver E)/                 # Driver E: Route Vw01-Vw17
│   │   └── Y (Driver D)/                  # Driver D: Route Y1, Y2
│   └── Uncategorised IOVNB Dataset/       # Flat dataset collection
│       ├── S-Dataset/                     # 70 Smartphone CSV files (176.32 MB, 1.34M rows)
│       └── V-Dataset/                     # 48 Vehicle ECU CSV files (202.52 MB, 1.49M rows)
└── Unsynchronised V and S Dataset/        # Independent single-sensor recordings
    ├── Categorised IOVNB (V) Dataset/     # ECU-only categorised recordings
    │   └── V Dataset/                     # 9 Vehicle ECU CSV files (101.20 MB, 0.50M rows)
    └── Uncategorised IOVNB (V and S) Dataset/# Independent flat recordings
        ├── S-Dataset/                     # 45 Smartphone CSV files (169.53 MB, 1.54M rows)
        └── V-Dataset/                     # 8 Vehicle ECU CSV files (91.18 MB, 0.49M rows)
```

Complete Dataset File & Volume Statistics

Dataset Directory Category	Sensor Channel Type	File Count	Total Rows	Disk Volume (MB)
Synchronised Categorised	Vehicle ECU (V)	47	1,493,590	202.52 MB
Synchronised Categorised	Smartphone (S)	44	1,343,921	168.30 MB
Synchronised Uncategorised	Vehicle ECU (V)	47	1,493,590	202.52 MB
Synchronised Uncategorised	Smartphone (S)	70	1,343,921	176.32 MB
Unsynchronised Categorised	Vehicle ECU (V)	9	502,408	101.20 MB
Unsynchronised Uncategorised	Vehicle ECU (V)	8	497,592	91.18 MB
Unsynchronised Uncategorised	Smartphone (S)	45	1,541,205	169.53 MB
TOTAL REPOSITORY SUMMARY		**273 CSVs**	**5,872,306**	**1,111.58 MB (~1.11 GB)**

2. Coordinate Systems & Coordinate Transformation Matrices

To combine inertial measurement unit (IMU) data, wheel speed odometry, and global navigation satellite systems (GNSS/GPS), IO-VNBD relies on three distinct reference frames and mathematical coordinate transformation matrices.

```
+------------------------+      Rotation R_s^b      +------------------------+
| Smartphone IMU Frame   | -----------------------> | Vehicle Body Frame (b) |
| (x_s, y_s, z_s)        |                          | (x_b, y_b, z_b)        |
+------------------------+                          +------------------------+
                                                                |
                                                                | Yaw Heading ψ_h
                                                                v
+------------------------+     ECEF-to-ENU R_ecef^enu +------------------------+
| WGS84 Geodetic (GPS)   | ------------------------> | Local Navigation (n)   |
| (φ, λ, h)               |                           | (East, North, Up)      |
+------------------------+                           +------------------------+
```

A. Global Geodetic Coordinate Frame (WGS84)

GPS data in both V- and S- datasets are reported in the World Geodetic System 1984 (WGS84) standard:

- Latitude (φ): Degrees North (Range: 51.946° to 53.793° in UK drives)
- Longitude (λ): Degrees East/West (Range: –2.254° to –0.603°)
- Altitude / Height (h): Kilometers in ECU (20.19 to 406.33 m converted) or meters in Smartphone (0 to 441.65 m)

Geodetic to Earth-Centered Earth-Fixed (ECEF) Transformation Matrix

To perform spatial distance or displacement calculations, latitude φ, longitude λ, and height h are transformed into ECEF cartesian coordinates $[X, Y, Z]^T$:

$$X_{ECEF} = (N(\varphi) + h)\cdot\cos\varphi\cdot\cos\lambda \quad Y_{ECEF} = (N(\varphi) + h)\cdot\cos\varphi\cdot\sin\lambda \quad Z_{ECEF} = (N(\varphi)(1 - e^2) + h)\cdot\sin\varphi$$

Where:

- $a = 6{,}378{,}137.0$ m (WGS84 semi-major axis)
- $e^2 = 6.69437999014 \times 10^{-3}$ (First eccentricity squared)
- $N(\varphi) = a / \sqrt{(1 - e^2 \sin^2\varphi)}$ (Prime vertical radius of curvature)

ECEF to Local East-North-Up (ENU) Navigation Frame Transformation Matrix

For vehicle dead-reckoning relative to a local origin $[\varphi_0, \lambda_0, h_0]^T$:

$$[E, N, U]^T = R_{ECEF}^{ENU} \cdot [X_{ECEF} - X_0,\ Y_{ECEF} - Y_0,\ Z_{ECEF} - Z_0]^T$$

where R_{ECEF}^{ENU} =

$$\begin{vmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{vmatrix}$$

B. Vehicle Body Coordinate Frame (b-Frame: Ford Fiesta ECU)

The vehicle body frame is defined according to ISO 8855 standards:

- X_b (Longitudinal Axis): Parallel to vehicle length, positive forward along travel direction
- Y_b (Lateral Axis): Horizontal axis perpendicular to X_b, positive to the right
- Z_b (Vertical Axis): Perpendicular to road plane, positive downwards or upwards

Body-to-Navigation Frame Rotation Matrix (R_b^n)

Using the vehicle's compass heading angle ψ_h (degrees from True North):

$R_b^n(\psi_h)$ =

$$\begin{vmatrix} \sin\psi_h & \cos\psi_h & 0 \\ \cos\psi_h & -\sin\psi_h & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Wheel Speed & Yaw Dynamics Kinematics Matrix

Vehicle forward velocity V_x and yaw rate $\dot{\psi}$ are linked to individual wheel angular speeds (ω_{FL}, ω_{FR}, ω_{RL}, ω_{RR} in rad/s) by:

$$v_{front\_left} = r_{wheel} \cdot \omega_{FL} \qquad v_{front\_right} = r_{wheel} \cdot \omega_{FR} \qquad v_{rear\_left} = r_{wheel} \cdot \omega_{RL} \qquad v_{rear\_right} = r_{wheel} \cdot \omega_{RR}$$

$$V_{vehicle} \approx (r_{wheel} / 4) \cdot (\omega_{FL} + \omega_{FR} + \omega_{RL} + \omega_{RR})$$

$$\dot{\psi}_{wheel} = (r_{wheel} / w_{track}) \cdot (\omega_{FR} - \omega_{FL})$$

Where r_{wheel} is the effective dynamic tire rolling radius (dependent on tire pressure categories A–E) and w_{track} is the wheel track width (1473–1493 mm for Ford Fiesta).

C. Smartphone Sensor Coordinate Frame (s-Frame) & Gravity Matrix

The smartphone's internal 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer operate in the smartphone body frame:

- Positive X_s: Points towards the right edge of the smartphone screen
- Positive Y_s: Points along the length of the smartphone screen towards the top (aligned with forward vehicle travel when mounted)
- Positive Z_s: Points outward, perpendicular to the screen surface

Smartphone-to-Vehicle Alignment Rotation Matrix (R_s^b)

When mounted in a phone holder attached to the vehicle windshield/dashboard, the orientation Euler angles (Roll φ_s, Pitch θ_s, Yaw ψ_s) form the 3D rotation matrix:

$$R_s^b(\varphi_s, \theta_s, \psi_s) = R_z(\psi_s) \cdot R_y(\theta_s) \cdot R_x(\varphi_s)$$

Expanded form, R_s^b =

$$\begin{vmatrix} \cos\psi_s \cos\theta_s & \cos\psi_s \sin\theta_s \sin\varphi_s - \sin\psi_s \cos\varphi_s & \cos\psi_s \sin\theta_s \cos\varphi_s + \sin\psi_s \sin\varphi_s \\ \sin\psi_s \cos\theta_s & \sin\psi_s \sin\theta_s \sin\varphi_s + \cos\psi_s \cos\varphi_s & \sin\psi_s \sin\theta_s \cos\varphi_s - \cos\psi_s \sin\varphi_s \\ -\sin\theta_s & \cos\theta_s \sin\varphi_s & \cos\theta_s \cos\varphi_s \end{vmatrix}$$

Gravity Vector Isolation & Compensation Matrix

Because vehicular vibrations and smartphone tilt introduce gravitational acceleration into total raw accelerometer measurements (a_{raw}), the dataset explicitly contains isolated gravity components (g_x, g_y, g_z):

$$a_{x,linear} = a_{x,raw} - g_x \quad a_{y,linear} = a_{y,raw} - g_y \quad a_{z,linear} = a_{z,raw} - g_z$$

where $\|\mathbf{g}\| = \sqrt{(g_x^2 + g_y^2 + g_z^2)} \approx 9.8066$ m/s²

3. Complete Telemetry Channel & Metric Inventory

The dataset comprises 53 distinct telemetry channels split across the Vehicle ECU logger and Smartphone sensors.

A. Vehicle ECU Telemetry Metrics (V- Dataset: 29 Channels)

#	Column Name	Unit	Rate	Min	Max	Mean	Description & Purpose
1	No of GPS Satellites Available	Count	10 Hz	0.0	140.0	116.24	GPS signal quality indicator & satellite constellation visibility.
2	Time Since Start of Day	Seconds	10 Hz	29,606.4	72,986.0	49,548.03	Epoch timestamp in seconds from UTC midnight.
3	Latitude	Degrees	10 Hz	51.9463°	53.7928°	52.5529°	WGS84 Geodetic Latitude from Racelogic VBOX GPS receiver.
4	Longitude	Degrees	10 Hz	−2.2535°	−0.6026°	−1.5186°	WGS84 Geodetic Longitude from Racelogic VBOX GPS receiver.
5	Velocity	km/h	10 Hz	0.0	131.88	43.81	True vehicle ground speed computed from GPS doppler shift.
6	Heading	Degrees	10 Hz	0.0°	360.0°	183.41°	Vehicle azimuth orientation angle relative to True North.
7	Height	km	10 Hz	0.0202	0.5343	0.1139	Vehicle altitude above WGS84 ellipsoid (in km).
8	Vertical velocity	km/h	10 Hz	−15.95	+13.53	+0.0118	Rate of elevation change (upward/ downward velocity).
9	Sample period	Seconds	10 Hz	0.0990	0.2000	0.1000	Nominal time delta Δt between rows (100 ms = 10 Hz).
10	Steering Angle	Degrees	10 Hz	0.0°	800.8°	59.22°	Steering wheel angular position from CAN Bus encoder.
11	Wheel Speed Front Left	rad/s	10 Hz	0.0	131.33	43.99	Front-left wheel rotational speed ω_{FL}.
12	Wheel Speed Front Right	rad/s	10 Hz	0.0	131.47	43.95	Front-right wheel rotational speed ω_{FR}.
13	Wheel Speed Rear Left	rad/s	10 Hz	0.0	131.66	43.93	Rear-left wheel rotational speed ω_{RL}.
14	Wheel Speed Rear Right	rad/s	10 Hz	0.0	130.81	43.75	Rear-right wheel rotational speed ω_{RR}.
15	Yaw Rate	deg/s	10 Hz	−66.0°	+66.6°	−0.353°	Angular velocity around vehicle vertical body axis (Z_b).
16	Indicated Vehicle Speed	km/h	10 Hz	0.0	131.36	43.97	Dashboard speed readout extracted from ECU wheel encoders.

17	Indicated Longitudinal Accel.	g	10 Hz	−1.0108	+0.4804	−0.0009	Forward/backward acceleration a_x (1g ≈ 9.81 m/s²).
18	Indicated Lateral Acceleration	g	10 Hz	−0.9007	+0.8701	−0.0031	Cornering/side-to-side lateral acceleration a_y.
19	Handbrake	Binary	10 Hz	0	1	0.0304	Handbrake engagement state flag (0 = off, 1 = engaged).
20	Gear Requested	Index	10 Hz	0	14	4.02	Transmission target gear selection request.
21	Gear	Index	10 Hz	0	5	2.88	Actual engaged mechanical transmission gear (1 to 5).
22	Engine Speed	RPM	10 Hz	0.0	6,332.0	1,651.92	Engine crankshaft rotation speed in revolutions per minute.
23	Coolant Temperature	°C	10 Hz	0.0°	105.0°	87.91°	Engine cooling fluid temperature.
24	Clutch Position	Binary	10 Hz	0	0	0.0000	Clutch pedal depress state flag (0 = released).
25	Brake Pressure	PSI	10 Hz	−1.91	+192.85	2.48	Hydraulic brake line fluid pressure.
26	Brake Position	Binary	10 Hz	0	1	0.2307	Mechanical brake pedal state flag (0 = released, 1 = pressed).
27	Battery Voltage	Volts	10 Hz	0.0	14.60	13.93	ECU electrical system voltage.
28	Air Temperature	°C	10 Hz	0.0°	30.75°	12.79°	Ambient outside air temperature.
29	Accelerator Pedal Position	%	10 Hz	0.0%	99.0%	9.57%	Throttle pedal depressing percentage (0 to 100%).

B. Smartphone Sensor Metrics (S- Dataset: 24 Channels)

#	Column Name	Unit	Rate	Min	Max	Mean	Description & Purpose
1	GPS LATITUDE	Degrees	1 Hz	51.9464°	53.7928°	52.5748°	Smartphone GPS latitude coordinate (WGS84).
2	GPS LONGITUDE	Degrees	1 Hz	−2.2535°	−0.6027°	−1.5453°	Smartphone GPS longitude coordinate (WGS84).
3	GPS ALTITUDE	Meters	1 Hz	0.0	441.65	167.80	Smartphone GPS altitude above mean sea level.
4	GPS SPEED	km/h	1 Hz	−0.18	+35.81	12.45	Vehicle speed derived from smartphone GPS module.
5	GPS ACCURACY	Meters	1 Hz	1.0	2,400.0	15.77	Estimated 2D GPS horizontal position accuracy radius.

6	GPS ORIENTATION	Degrees	1 Hz	0.0°	360.0°	178.39°	Heading direction of motion from smartphone GPS.
7	GPS SATELLITES IN RANGE	Text	1 Hz	"0 / 0"	"24 / 28"	"18 / 19"	Tracked vs in-view GPS satellite ratio.
8	TIME SINCE START	ms	10 Hz	5.0	14,502,710	3,425,306	Elapsed recording time in milliseconds.
9	DATE	Timestamp	10 Hz	ISO 8601	ISO 8601	N/A	High-resolution timestamp YYYY-MM-DD HH:MI:SS_SSS.
10	ACCELEROMETER X	m/s²	10 Hz	−49.20	+26.12	−0.0017	Total linear + gravity acceleration along lateral axis X_s.
11	ACCELEROMETER Y	m/s²	10 Hz	−39.36	+52.40	−0.0362	Total linear + gravity acceleration along longitudinal axis Y_s.
12	ACCELEROMETER Z	m/s²	10 Hz	−28.07	+59.45	+9.8461	Total linear + gravity acceleration along vertical axis Z_s.
13	GRAVITY X	m/s²	10 Hz	−4.639	+2.391	+0.00002	Isolated gravity acceleration component along X_s.
14	GRAVITY Y	m/s²	10 Hz	−6.367	+4.869	−0.00017	Isolated gravity acceleration component along Y_s.
15	GRAVITY Z	m/s²	10 Hz	+7.063	+9.807	+9.8065	Isolated gravity acceleration component along Z_s.
16	GYROSCOPE Yaw	rad/s	10 Hz	−9.811	+13.692	−0.00036	Gyroscope angular velocity around Z_s axis.
17	GYROSCOPE Pitch	rad/s	10 Hz	−10.873	+13.912	−0.00280	Gyroscope angular velocity around X_s axis.
18	GYROSCOPE Roll	rad/s	10 Hz	−7.192	+12.872	+0.00011	Gyroscope angular velocity around Y_s axis.
19	MAGNETIC FIELD X	µT	10 Hz	−107.50	+58.00	−10.49	Geomagnetic flux density along X_s in microteslas.
20	MAGNETIC FIELD Y	µT	10 Hz	−170.75	+53.46	−33.09	Geomagnetic flux density along Y_s in microteslas.
21	MAGNETIC FIELD Z	µT	10 Hz	−64.62	+159.44	+16.33	Geomagnetic flux density along Z_s in microteslas.
22	ORIENTATION (Yaw)	Degrees	10 Hz	0.0°	360.0°	135.48°	Computed Euler orientation Yaw angle ψ_s.
23	ORIENTATION (Pitch)	Degrees	10 Hz	−90.0°	+84.27°	−83.29°	Computed Euler orientation Pitch angle θ_s.
24	ORIENTATION (Roll)	Degrees		−180.0°	+180.0°	−66.05°	

	10 Hz	Computed Euler orientation Roll angle φ_s.

4. Experimental Design Matrices

A. Tyre Pressure Matrix (Table 5 in Paper)

Variations in tire inflation pressures directly impact tire rolling radius r_{wheel}, cornering stiffness, and vibration frequencies:

Notation Code	Front Right	Front Left	Rear Right	Rear Left	Experimental Purpose
A	16 PSI	15 PSI	14 PSI	14 PSI	Severely under-inflated tire dynamics test.
B	31 PSI	31 PSI	25 PSI	25 PSI	Asymmetric manufacturer recommended specification.
C	33 PSI	33 PSI	31 PSI	27 PSI	Asymmetric load distribution setting.
D	33 PSI	33 PSI	26 PSI	26 PSI	Standard nominal vehicle manufacturer pressure.
E	N/A	N/A	N/A	N/A	Unmeasured baseline / secondary vehicle setup.

B. Driver Profiles & Driving Style Matrix (Table 1 & Tables A1–A7)

Driver	Driving Style	Primary Vehicle	Recording Locations	Miles / Hours
Driver A	Defensive	Ford Fiesta Titanium	England (Coventry, Rugby, Nuneaton)	~350 km / ~10.2 hrs
Driver B	Defensive	Ford Fiesta Titanium	England (Coventry Ring Roads)	~105 km / ~3.0 hrs
Driver C	Defensive	Ford Fiesta Titanium	England (Warwick, Oxford, Chesterton)	~280 km / ~5.8 hrs
Driver D	Defensive	Ford Fiesta Titanium	England (Coventry, Kenilworth)	~108 km / ~3.5 hrs
Driver E	Aggressive	Ford Fiesta Titanium	England (Burton, Buxton, Bradford)	~650 km / ~18.5 hrs
Driver F	Defensive	Renault Megane	France	~1,517 km / ~16.8 hrs
Driver G	Defensive	Toyota Corolla Verso	Nigeria	~0.06 km / ~0.16 hrs
Driver H	Defensive	Volvo XC70	England	~1,512 km / ~10.6 hrs

C. 32 Driving & Environmental Scenarios Matrix (Table 2)

1. Hard Brake
2. Sharp Turn Left and Right
3. Swift Maneuvers
4. Round-about (x1 to x77)
5. Rain & Wet Roads
6. Night and Day Driving
7. Skid Maneuvers
8. Mountain & Hilly Roads
9. Dirt & Gravel Roads
10. Country Roads
11. Motorway (M1, M6, M40, M42)
12. Town Centre Driving
13. Traffic Congestion
14. Successive Left/Right Turns

15. Varying Short Accelerations
16. A-Roads (A45, A50, A515)
17. B-Roads (B4101, B5053)
18. Wet & Muddy Roads
19. U-turns / Reverse Drives
20. Mud Road
21. Varying Tyre Pressure (A-E)
22. Vehicle Drifts
23. Bumps
24. Inner City Driving
25. Winding Roads
26. Zig-Zag Drives
27. Approx. Straight Motion
28. Parking Navigation
29. Potholes
30. Residential Roads
31. Stationary (Bias Calibration)
32. Valley Driving

5. Benchmark Applications & Machine Learning Formulation

IO-VNBD serves as a foundational dataset for training and testing data-driven deep learning models for autonomous vehicle localization:

Inertial Odometry & Dead Reckoning

Training Recurrent Neural Networks (RNNs, LSTMs), Convolutional Neural Networks (CNNs), and IONet architectures to regress displacement (Δx, Δy) and heading change $\Delta\psi$ directly from 10 Hz IMU acceleration and gyro sequences during GPS outages:

$$\hat{s}_{k+1} = f_\theta(a_{k-N:k}, \omega_{k-N:k})$$

Sensor Bias & Noise Mitigation

Using the >20 minutes of Stationary dataset recordings (V-Vw1, V-Vw15) to isolate zero-velocity update (ZUPT) biases in low-cost MEMS accelerometers and gyroscopes:

$$b_{accel} = \mathbb{E}[a_{stationary} - g] \qquad b_{gyro} = \mathbb{E}[\omega_{stationary}]$$

Wheel Encoder & Inertial Fusion (EKF / UKF)

Fusing 4-wheel rotational speeds (ω_{FL}, ω_{FR}, ω_{RL}, ω_{RR}) with steering angle δ and smartphone IMU to eliminate longitudinal wheel slip and lateral sideslip angle β.

Verification & Summary Index

Total CSV Files Verified	**273 files**
Total Data Volume	**1,111.58 MB (~1.11 GB)**
Total Telemetry Samples	**5,872,306 rows**
Documented Channels	**29 Vehicle ECU + 24 Smartphone = 53 total metrics**
Coordinate Frameworks	**WGS84 Geodetic, ECEF, Local ENU, Body b-frame, Smartphone s-frame**